Generalization. Out of Sample Performance.

Big Data y Machine Learning para Economía Aplicada

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Universidad de los Andes

Agenda

1. Previously ...
2. Generalization
   - Overfitting and Underfitting
   - Out-of-Sample Error Estimation
     - AIC: Akaike Information Criterion
     - SIC/BIC: Schwarz/Bayesian Information Criterion
   - Cross-Validation
3. Review

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Set up

- We observe (Y,X)
- We choose a predictor $f : \mathcal{X} \to \mathbb{R}$ to minimize expected loss.

$$f^*(X) = \underset{f}{\operatorname{argmin}}\, R(f(X)) \tag{1}$$

$$= \underset{f}{\operatorname{argmin}}\, \mathbb{E}\left[(Y - f(X))^2\right]. \tag{2}$$

Linear model

$$f(X) = Xb \tag{3}$$

Uncertainty

- Parameters
- Model

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Generalization Overview

- In ML we care in out-of-sample prediction
- Generalization refers to a model's performance on unseen data.

Generalization

- Unseen data is typically refered as **test data**,
- While the sample data is called the **training data**.
- Objective:

$$f^*(X) = \underset{f}{\operatorname{argmin}}\, \mathbb{E}\left[(Y - f(X))^2\right]. \tag{4}$$

Linear model

Again...

$$f(X) = Xb \tag{5}$$

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Generalization

Overfitting and Underfitting

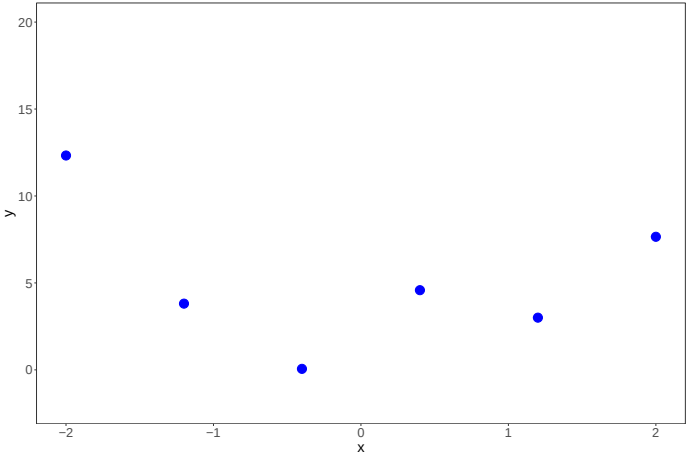


Figure 1: Training Data

Generalization

Overfitting and Underfitting

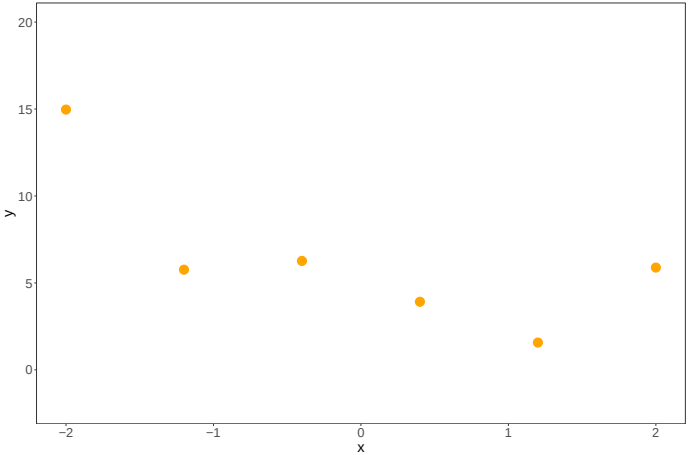
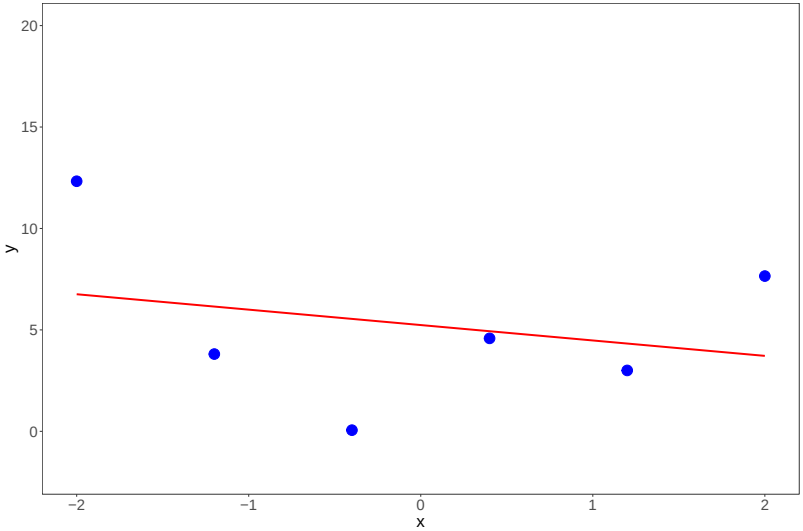


Figure 2: Testing (Unseen) Data

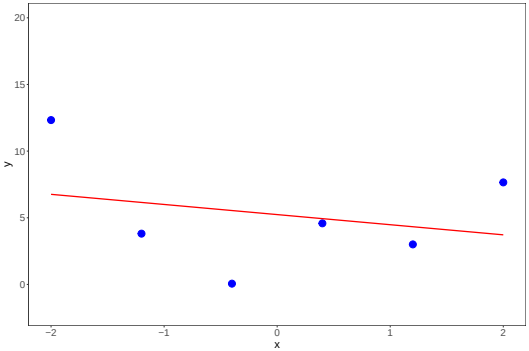
Generalization

Overfitting and Underfitting

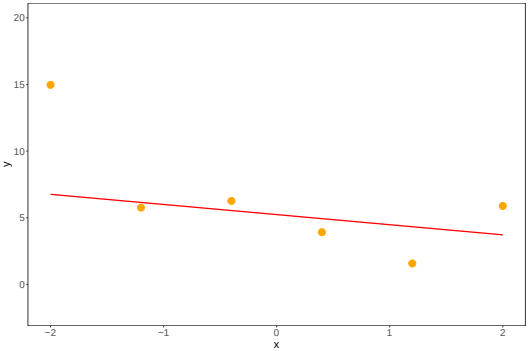


Generalization

Out-of-Sample Performance



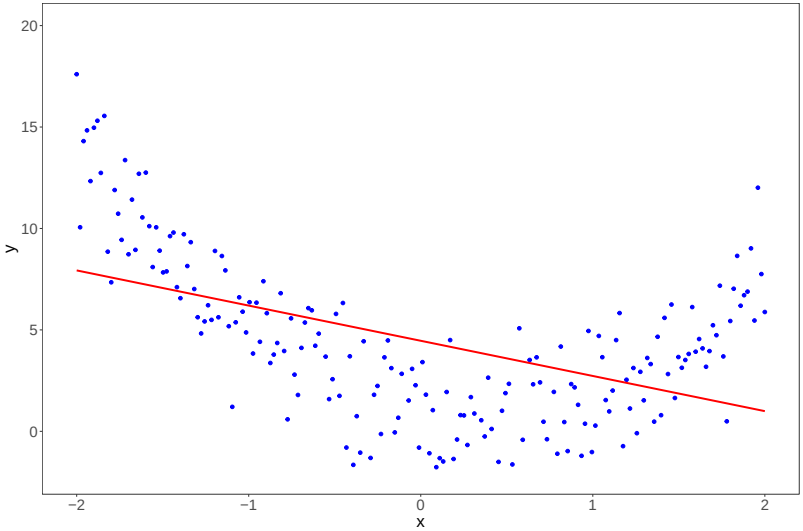
(a) Training Data



(b) Testing Data

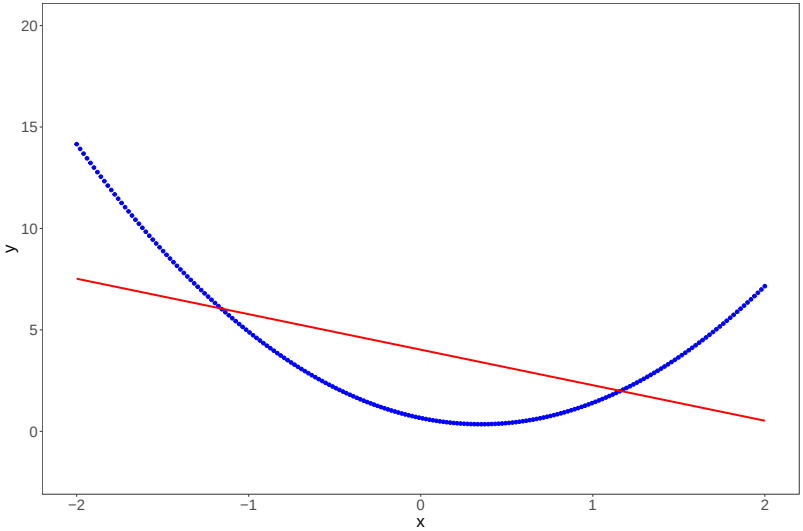
Generalization

Out-of-Sample Performance. More data?



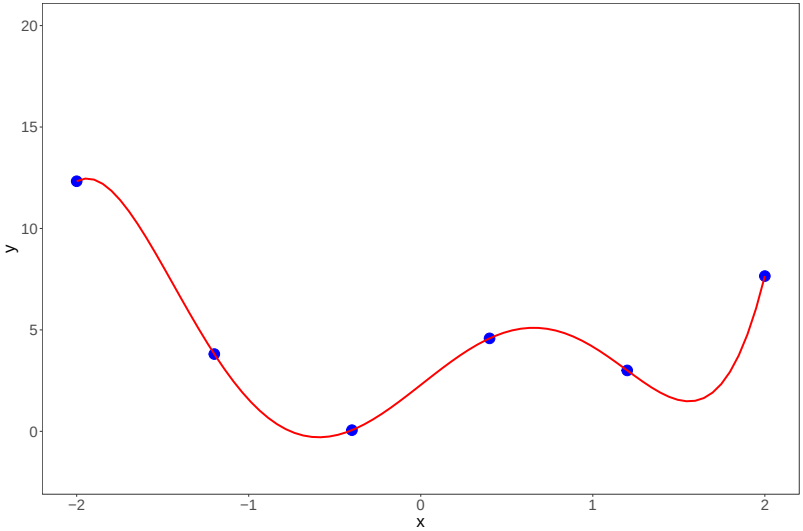
Generalization

Out-of-Sample Performance. Noiseless data?



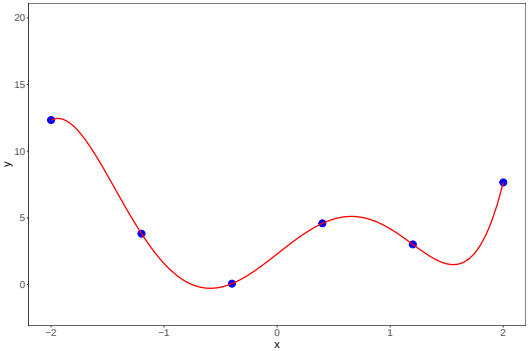
Generalization

More Complex Model

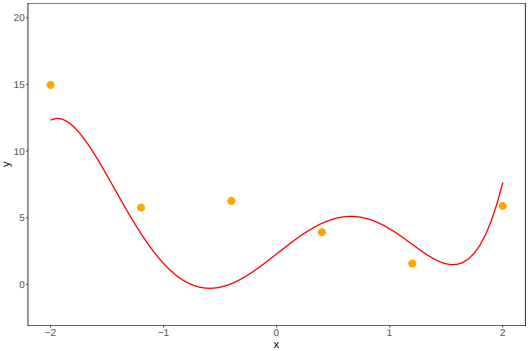


Generalization

More Complex Model. Out-of-Sample Performance



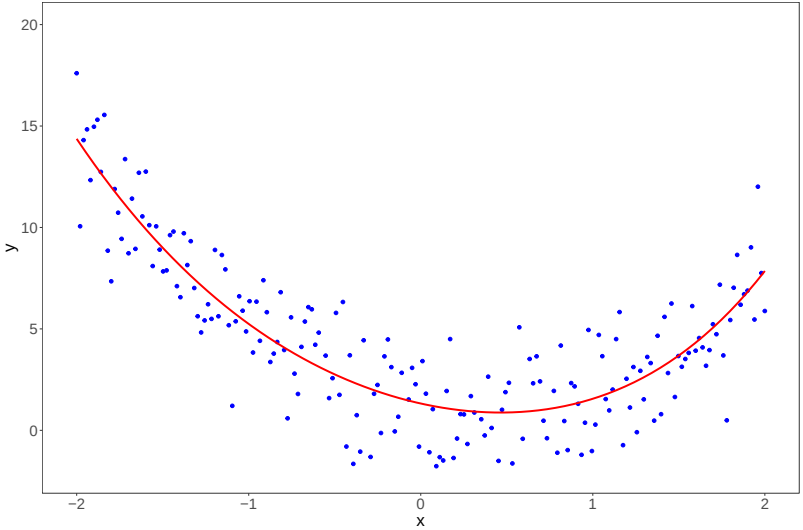
(a) Training Data



(b) Testing Data

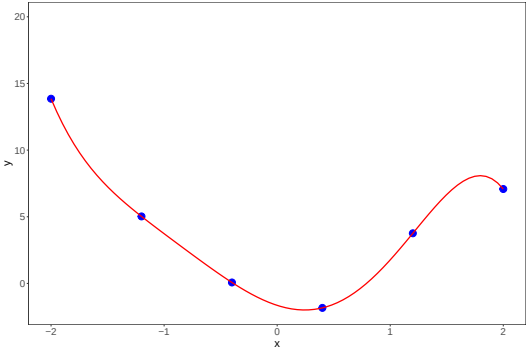
Generalization

More Complex Model. Out-of-Sample Performance: More Data

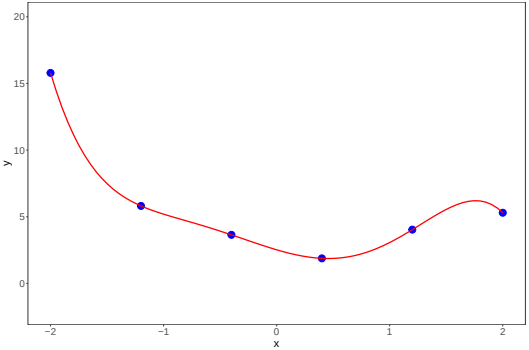


Generalization

More Complex Model. Out-of-Sample Performance: Variance



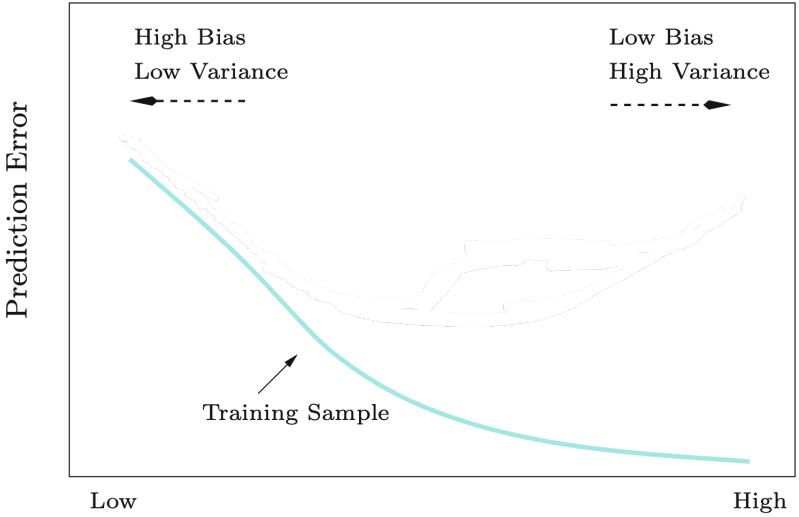
(a) Training Data 2



(b) Training Data 3

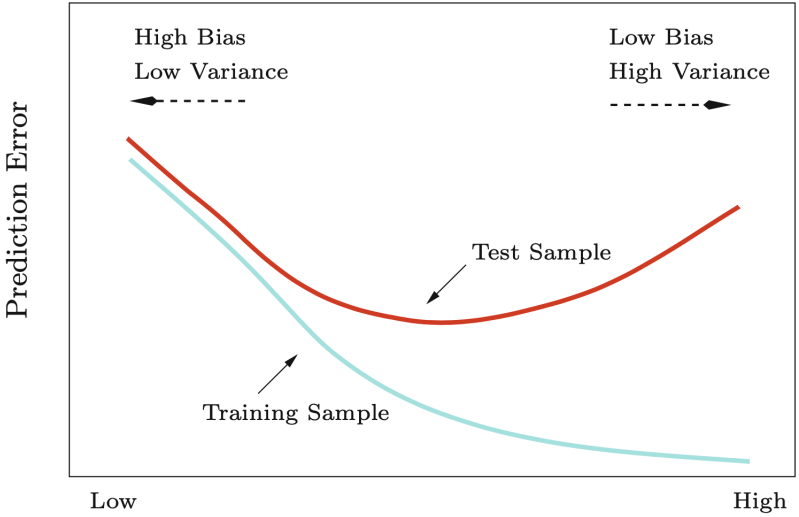
Generalization

Overfitting and Underfitting. Bias-Variance Tradeoff



Generalization

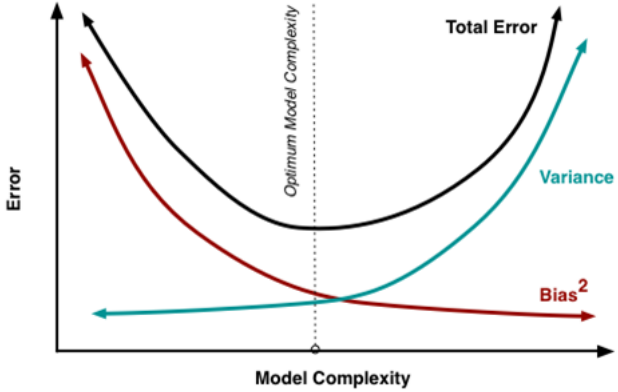
Overfitting and Underfitting. Bias-Variance Tradeoff



Mathematical Decomposition for Regression

Mathematical Decomposition for Regression

Bias-Variance Tradeoff



Source: https://tinyurl.com/y4lvjxpc

- ML best kept secret: By tolerating some bias we can have significant gains in variance

Generalization

Key Insights on Bias-Variance Tradeoff

- The bias term reflects the error introduced by the model's inability to approximate the true function f^*.
- The variance term reflects the sensitivity of the model to the specific training set.
- As dataset size increases, variance generally decreases.
- The noise term σ^2 is unavoidable and cannot be predicted.
- The decomposition for classification problems is less clear than for regression problems, but still present.

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Generalization

Train and Test Sets. In-Sample and Out-of-Sample Prediction.

- Como seleccionamos la parametrización que minimize el error de predicción fuera de muestra?
- Problema: solo contamos con una muestra

Generalization Error

- Para seleccionar la mejor parametrización con respecto al Test Error (error de prueba), es necesario estimarlo.
- Hay dos enfoques comunes:
  - Podemos estimar indirectamente el error de la prueba haciendo un ajuste al error de entrenamiento para tener en cuenta el sesgo debido al sobreajuste $\Rightarrow$ Penzalización ex post: AIC, BIC, MSE_{adj}, R^2_{adj}, etc.

Test Error

AIC

- Akaike (1969) fue el primero en ofrecer un enfoque unificado al problema de la selección de modelos.
- Elegir el modelo j tal que se minimice:

$$AIC(j) = log\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2\right) - p_j \tag{6}$$

Test Error

SIC/BIC

- Schwarz (1978) mostró que el AIC es inconsistente, (cuando $n \to \infty$, tiende a elegir un modelo demasiado grande con probabilidad positiva)
- Schwarz (1978) propuso:

$$SIC(j) = log\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2\right) - \frac{1}{2}p_j log(n) \tag{7}$$

Test Error

AIC vs BIC

$$AIC(j) = log\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2\right) - p_j \tag{8}$$

$$SIC(j) = log\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2\right) - p_j\frac{1}{2}log(n) \tag{9}$$

Test Error

MSE_{adj}, R^2_{adj}

Generalization Error

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- Hay dos enfoques comunes:
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  - Usar la misma muestra y estimar directamente el Test Error

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Test Error

Cross-Validation

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

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Review

Hoy

- Bias-Variance Tradeoff (Dilema Sesgo/Varianza)
- Sobreajuste y Selección de modelos
  - AIC, BIC, R^2_{adj}
  - Enfoque de Validación
  - LOOCV
  - K-fold Cross-Validation (Validación Cruzada)